

Demographic vs Behaviour -Based Customer Segmentation: A Stability-Tested, Externally Validated Comparison

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Abstract

Customer segmentation helps companies market better by grouping customers who act or look similar.. How customers are represented can change how good, steady and useful the groups are. This study looked at three ways to represent customers: just using details using how they behave and a mix of both. The data used had 2,240 entries with 29 pieces of information. After cleaning 2,232 entries were left. The data was split into three types: details with five pieces of information behavior with seven and a mix with 12. The data was made consistent and missing parts were filled in. Then two methods of grouping were tested: K-Means and Wards hierarchical method. These were checked using three scores: silhouette, Davies–Bouldin and Calinski–Harabasz. The study also looked at how steady the groups were by trying starting points using only 80% of the data changing the features and checking against real results from the campaign that were not used for grouping. At three groups the behavior-based K-Means had a silhouette score of 0.319 and a Davies–Bouldin score of 1.434. These were better than the basic and mix versions on these tests. The behavior-based K-Means was very steady when tested with 100 samples with an average score of 0.977. The mix version was also very steady when tested with starts and samples. The basic grouping had the difference in how customers responded to the campaign with 18.66 percentage points between the groups. The behavior and mix groupings explained more of the differences in how much customers spent with scores of 0.675 and 0.685. The results show that behavior is a main way to group customers for marketing while basic details add extra understanding for targeting. These results are, for this data set and should not be seen as proof that marketing causes changes. The data source and three key papers need to be checked before the study is finalized.

Keywords: customer segmentation; K-Means; hierarchical clustering; cluster stability; behavioural analytics; campaign response.

1. Introduction

Customer segmentation takes a group of customers that're different from each other and divides them into smaller groups that are more alike within themselves. These groups can be helpful for making decisions about marketing strategies. This method is often used to plan marketing campaigns manage relationships with customers choose the channels and decide how to spend marketing money. Algorithmic segmentation allows using a lot of customer details together.. The results depend on how the customers are described, which details are chosen, the method used for grouping and the rules used to decide what makes a good grouping.

A common problem in how segmentation's used is that people pick a method because it gives a good score on one specific measure. This can miss whether the groups can be made again if different details lead to groups and if the groups are different when it comes to business results that were not used to make the groups. Just because a group is separated in a way does not mean it is useful for business.

This project looks at three ways of describing customers: just based on age, gender and location based on what they do and a mix of both. It uses two grouping methods K-Means and Ward hierarchical clustering. It checks how good the groups are inside how consistent they are when started again with settings and different parts of the data how much the groups match each other and how well they separate customers based on results from real marketing campaigns and spending. The questions this research asks are: (1) How do the different ways of describing customers compare when looking at how good the groupsre inside? (2) How consistent are the groups when started in ways and with different data parts? (3) How much do the different ways of describing customers. What details make the groups different? (4) Which way of describing customers separates customers better when it comes to results from marketing campaigns and spending?

The main goal is to provide a way of looking at things that's useful for business and not just new technology. It looks at how good the groupsre how steady they are and how well they show business differences. This matches what some experts have said about using than one way to check results instead of just one score or one method (Hennig, 2015; Wedel & Kannan 2016; the review by Hutter et al., 2023).

Three earlier studies help explain where this research fits in: a review of how to use algorithms for customer segmentation (Salminen et al., 2023) a look at how to segment customers for shops (Gomes & Meisen 2023) and a study, about how stable customer groups are (Müller & Hamm 2014). A recent study that used RFM and K-Means with a marketing campaign dataset gives another point of comparison (Naufal & Alamsyah 2026).

2. Literature Review

2.1 Segmentation as a decision problem

Segmentation is useful when the groups that are created can help make a decision. Marketing analytics links customer data to decisions like who to contact what offer to make and which channel to use (Wedel & Kannan 2016). Customer-level data can help with targeting. A segmentation needs to be easy to understand and useful for action instead of just being mathematically simple. Market segmentation texts say that segmentation choices depend on the market the data that's available and how the segmentation will be used by managers (Wedel & Kamakura 2000; Dolnicar et al. 2018).

The implication, for this study is that cluster quality should be checked in than one way. Internal metrics look at the shape and structure of the groups. Stability tests check if similar data produce groups. External validation checks if the groups separate variables that matter to a business but were not used in the clustering process. These methods answer questions and might support different groupings.

2.2 K-Means and hierarchical clustering

K-Means divides observations into a number of groups by reducing the sum of squared distances inside each cluster. K-Means is fast easy to use. Gives clear cluster profiles.. K-Means reacts strongly to the first centroids, the scaling of variables and to outliers. K-Means also likes clusters that're tight and roughly spherical in the feature space that you choose. When you change how the features are represented the K-Means results can shift. Therefore comparing data to behavioural data is not a neutral step; it alters the distances that K-Means uses.

Hierarchical clustering creates a tree of groups. Ward linkage joins groups so that the rise in within-cluster variance is as small, as possible. You can look at the tree at levels, which helps when you try to understand the data.. To get a flat set of clusters you still have to choose a cut level. Hierarchical methods also tend to be slower. Can react to the choice of distance and linkage rules. For a comparison it is best to use the same number of clusters so that we can compare how different representations behave. Still the ideal number of clusters may be different if you pick a rule.

2.3 Internal cluster validation

The silhouette coefficient looks at how a customer is from the centre of its own cluster compared to how far it is from the nearest other cluster. A higher silhouette coefficient means the customer fits nicely in its cluster and the clustersre well separated (Rousseeuw, 1987). The Davies–Bouldin index mixes how spread out customers are inside a cluster with how apart clusters are. A lower Davies–Bouldin index is better (Davies & Bouldin 1979). The Calinski–Harabasz index measures the spread between clusters against the spread inside clusters. A higher Calinski–Harabasz index is better (Caliński & Harabasz 1974). No single index can say that a segmentation is absolutely correct. Each index has its

geometric preference and can give different rankings especially when cluster sizes or shapes are not the same (Halkidi et al., 2001; Arbelaitz et al., 2013).

In this analysis we show all three measures. We do not treat these measures as business value. That distinction is key when interpreting the results. A behavioural feature set can create geometric clusters but it might not give the biggest spread, in response rates. On the hand a demographic partition might have looser geometry but can separate response rates more strongly.

2.4 Stability and reproducibility

A stable segmentation should not rely much on a random starting point or a small change in the set of customers selected. The adjusted Rand index (ARI) is a way to measure how similar two groupings are while taking into account what might happen by chance (Hubert & Arabie 1985). In this study the ARI is used to compare K-Means runs to compare groupings from smaller samples with the main groupings on the same group of customers and to check how well different ways of representing the data agree with each other. Stability is not the same as correctness. A method can be stable. Still produce a poor grouping. That is why stability should be looked at together with measures—both internal and external.

Resampling-based validation is especially important when customer segments are meant to support business decisions. If the segmentation shifts a lot from one run, to the next it can lead to targeting. Campaigns may become hard to compare and unnecessary costs can rise from repeated rework. It is better to report the stability and also show how much it varies, rather than just showing the best single result.

2.5 Behavioural and demographic representations

Customer details include things like age, money, school, family setup. If someone is married. Action details include things like how much someone spends, where they buy from if they care about sales and how often they come in. Demographics can help explain who the customer is and managers might find them easy to get.. Just because people are similar in these areas doesn’t mean they act the same when buying. Action details are closer to what customers do and how they engage.. These can only show what happened before and might not predict what will happen next.

Recent looks at computer methods for grouping customers show that K-Means is often used.. There is a lot of different methods and ways to check them. Salminen and others looked at 172 papers. Say there should be more planned ways to check. Another look at grouping for shopping also shows K-Means is popular.. There is no one best way that works for all situations. These looks suggest that it is better to check than one way to group and more, than one way to measure.

2.6 Research gap and study positioning

The question here isn't just whether K-Means or Ward clustering is better overall. It's whether the way we represent customers. Using kinds of features. Changes how good, stable and useful the clusters are on the same dataset.

The notebook goes beyond a model comparison. It includes:

(a) behavioural and combined feature representations;

(b) several internal evaluation metrics;

(c) checks for stability using different random seeds and subsamples;

(d) measures of partition agreement;

(e) robustness tests; and

(f) external validation using response and spending variables that weren't used when forming the clusters.

This study is meant as a follow-up to earlier work not as a new algorithmic breakthrough. Salminen et al. (2023) Point out that there's a lot of variation in segmentation methods and evaluation tools. Gomes and Meisen (2023) stress that how we represent customers and how we evaluate results both matter. And that no one method fits every e-commerce situation. Naufal and Alamsyah (2026) did a RFM/K-Means analysis using a well-known Marketing Campaign dataset.

What's different here is that this analysis directly compares behavioural and combined features. It looks at both K-Means and Ward clustering. It also adds stability checks. Running the model times with different seeds and subsamples. And tests how well the clusters predict actual campaign responses and spending.

Since the source and version of the CSV file aren't documented in the provided materials any similarity, to the dataset used by Naufal and Alamsyah is just a context clue not proof the files are the same.

2.7 From statistical separation to actionable segments

A useful segmentation should be checked from data to the manager's decision. First features must show differences between customers. Second the algorithm must create groups that're clear and distinct. Third those groups must be easy for a marketing team to understand. Fourth the team must pick an action for each group and see if the action changes results. These stages must not be turned into a score. The current study looks at the two stages with feature comparison internal indices and stability tests. It also gives a look at the third and fourth stages using cluster profiles and differences in response or spending. A later plan should test if targeting by segments gives results than a campaign that does not use segments.

This difference matters because a segment can look different in numbers but still not help in practice. If two groups get the offer on the same channel segmentation can make things more complicated without benefit. On the hand a small cluster can be useful if it spots a high-value group that needs something different or reacts strongly to a particular treatment. The Ward behavioural result shows this trade-off: it did well on indices but made the smallest cluster only 9.41 % of the cleaned sample. The right move is not to dismiss the cluster away but to look at its profile its size, the cost to serve it and the possible value, to a campaign before putting it into use.

2.8 Synthesis of the literature and positioning of the study

The literature we looked at points to three ideas. First the quality of segmentation depends on how we represent customers and on the algorithm we use. Gomes and Meisen (2023) say that representing customers is a step before clustering and they find that RFM features and K-Means are common in e-commerce studies. This pattern backs the decision in this study to compare feature representations instead of assuming one algorithm will always be best. It also encourages adding channel use and deal purchase behavior with spending because customer value alone may not show how a marketing team can reach a customer.

Second no single internal index can prove that a segmentation is useful. Salminen et al. (2023) Show that published customer-segmentation studies use different algorithms and evaluation measures. The lesson is not that every project must use every metric but that the evaluation should fit the decisions the segments will support. This study therefore reports silhouette, Davies–Bouldin and Calinski–Harabasz indices, treating them as geometric summaries instead of direct campaign effectiveness measures. Adding response-rate differences and segment profiles gives a business view though the observational design cannot prove that using the segments will improve results.

Third we must look at the reliability of segments directly. Müller and Hamm (2014) point out that internal and dynamic stability cannot be taken for granted in market segmentation. This notebook tests how sensitive the results are to starts and to repeated subsamples and it compares the partitions with the adjusted Rand index. This goes beyond a single-run clustering because instability could shift customer labels without a change in behaviour. But resampling stability does not equal stability over time: the data we have does not include a confirmed long-term record of segment memberships. Future work should run the segmentation on periods and track how segments move before using them in practice.

Naufal and Alamsyah (2026) give an example of using RFM and K-Means on a public Marketing Campaign dataset. Their study helps as a comparison but we do not know the exact source or version of the CSV we use here. Therefore this paper does not assume the two datasets are the same nor does it see file sizes and field names as proof that they come from the same source. This contribution is narrower and more solid: it compares three representations on the file we have reports resampling

validation and checks whether the representations split historical campaign responses. The results should be read as evidence, for this dataset and pipeline not as a ranking of segmentation methods.

3. Methodology

3.1 Data and provenance

The file provided, marketing_campaign.csv is a tab-separated file that has 2,240 customer records and 29 columns. It includes details about the age, gender and location of customers. It also shows how money they spend on different types of products. The file has information about how they buy things through different channels how many deals they take, how many times they visit the website and details about marketing campaigns. The notebook shows that there are 24 missing values in the Income column. There are no rows that're exactly the same when all columns are looked at.. There are 182 rows that are the same when the ID is not considered. There is a column that has IDs, for each customer. The exact original source of the file is not mentioned in the files provided. There is a dataset that has the same size and the same kind of information. It is called the Customer Personality Analysis/Marketing Campaign dataset.. The group needs to check the file that was actually downloaded to make sure it is the same. Do not say where the file came from without being sure. Write down the web address, the person or organization that made the dataset, the date when it was accessed and any rules or agreements that go with it after checking.

3.2 Cleaning and feature engineering

The notebook converts Dt_Customer into a date format. Uses the most recent date in the dataset as a reference point to calculate age and customer tenure. This is an assumption because the actual collection date is not documented. Eight records were removed based on predefined rules: three customers with ages over 100 years one with income that exceeds the fence by more than three times the interquartile range and four with marital-status labels that read "Absurd" or "YOLO." After these exclusions the final analysis sample includes 2,232 customers. The 24 missing values for Income are filled in during the feature-building process using the value.

The demographic feature set includes Age Income, a version of education a binary Partnered indicator and Total_Children. The behavioral feature set includes Recency, log-transformed total spending, deal purchases, web purchases, catalogue purchases, store purchases and monthly web visits. Together these form a combined set of 12 features. The six product-spend variables are summed to create Total_Spend, which is then transformed using log(1. Total_spend) to prepare it for clustering. Campaign response, campaign acceptances and complaints are not included in the clustering inputs; they are kept separate for validation purposes.

Education is converted to a scale. The notebook treats "2n Cycle" as equivalent, to "Master," which's an assumption that should be checked or explained. After the imputation StandardScaler is applied to the features. The analysis uses a fixed seed of 42 to ensure reproducibility. PCA is considered as a tool to assess dimensionality and sensitivity. It is not automatically applied to all main models.

3.3 Clustering design and evaluation

K-Means and agglomerative hierarchical clustering with Ward linkage are tested. The main comparison that is reported sets k equal to 3 for behavioural and combined feature sets. This allows for a matched-k comparison. The notebook also has a demographic sensitivity analysis using the cluster count that is selected by the rule. This difference is important because the demographic silhouette increases when k increases. It does not show a natural elbow according to the projects initial rule. The main three-cluster comparison is therefore a choice made for comparability. It is not a claim that k equals 3 is the best option.

Internal evaluation uses silhouette, Davies–Bouldin and Calinski–Harabasz scores. Stability is checked with 30 K-Means initialisations. These initialisations use seeds from 0 to 29. Stability is also checked with 100 80% subsamples for K-Means and Ward. Additional stability comparisons are done across k values. ARI and normalized information are used to check how much agreement there is between different segmentations. Robustness tests include removing records while ignoring ID. They also include dropping records, with missing income of filling in the missing values. They use PCA that keeps 95% of the variance. They replace spend with six log-transformed spend categories. They also exclude education.

3.4 External validation and interpretation

External validation uses campaign Response, Campaigns_Accepted and Total_Spend while campaign variables are kept out of clustering. The notebook looks at how the response-rate spreads between clusters measures Cramér's V to see how strongly cluster and response are linked and checks eta-squared to compare differences, in campaign acceptances and spending. Bootstrap confidence intervals are also added for Cramér's V. Because the data is old records that were observed these links cannot prove that putting customers into a segment will cause them to respond more.

3.5 Reproducibility

At the time of running the notebook shows that Python 3.12.3 is used, along with NumPy 2.4.4 pandas 3.0.2, SciPy 1.17.1 scikit-learn 1.8.0, Matplotlib 3.10.8 and seaborn 0.13.2. These versions come from the existing notebook. You should not assume they exist on every computer. The repository that you submit must contain a requirements file and clear instructions, on how to run it. The notebook must be executed from start to finish in the intended environment before you finalize the submission.

3.6 Analytical choices and their implications

Standardisation is important for distance-based methods when the input variables are on scales. If not standardised variables like income or spending amounts might overpower variables that are measured as counts or binary indicators. Median imputation keeps the sample size the same. It reduces the variation in the imputed records compared to the original observed distribution. The missing-income

sensitivity check helps confirm that removing those records does not significantly change the results of the partitions.

The duplicate-record check is also important. Records that match on all customer attributes except for the ID could represent customers who are similar repeated entries or a feature of how the data was built. Removing these duplicates changes the sample size so this step should be seen as a sensitivity analysis, not a fix.

4. Results

4.1 Main internal comparison

The main comparison uses three clusters for each representation. Shows results from both K-Means and Ward hierarchical clustering. For K-Means behavioural segmentation got the silhouette score at 0.319, the lowest Davies–Bouldin index at 1.434 and the highest Calinski–Harabasz index at 995.730, out of the three representations. Demographic K-Means had a silhouette of 0.217 a Davies–Bouldin index of 1.606 and a Calinski–Harabasz index of 538.257. Combined K-Means scored a silhouette of 0.207 a Davies–Bouldin index of 1.830 and a Calinski–Harabasz index of 623.929.

Representation	**Algorithm**	**Silhouette ↑**	**Davies–Bouldin ↓**	**Calinski–Harabasz ↑**	**Smallest cluster**
Demographic	K-Means	0.217	1.606	538.3	27.0%
Demographic	Ward	0.183	1.767	467.0	22.1%
Behavioural	K-Means	0.319	1.434	995.7	24.9%
Behavioural	Ward	0.324	1.236	871.3	9.4%
Combined	K-Means	0.207	1.830	623.9	26.7%
Combined	Ward	0.193	1.881	575.8	21.1%

Ward clustering showed the overall pattern for silhouette and Davies–Bouldin. Behavioural Ward had a silhouette of 0.324 and Davies–Bouldin index of 1.236. These were the values, among all Ward solutions. However the smallest cluster represented 9.41% of customers. That is 210 people. This is a consideration. The internal metrics therefore support the representation.. Cluster size balance and interpretability should also be taken into account.

4.2 Stability

Behavioural K-Means had an ARI of 0.884 over 30 single-initialisation runs. It also had an ARI of 0.977 over 100 80% subsamples. Combined K-Means was more stable when it came to initialisations. Its average ARI was 0.987. It was also stable when it came to subsamples. Its average ARI was 0.976. Demographic K-Means had an average ARI over initialisations. Its average ARI was 0.445. However its 80% subsample ARI was 0.833. Ward subsample stability was lower than K-Means for all three representations. The ARI was 0.616 for demographic. The ARI was 0.607 for behavioural. The ARI was 0.738, for combined.

Representation	Stability test	Mean ARI	SD
Demographic	K-Means seed runs	0.445	0.289
Demographic	K-Means 80% subsamples	0.833	0.157
Demographic	Ward 80% subsamples	0.616	0.204
Behavioural	K-Means seed runs	0.884	0.195
Behavioural	K-Means 80% subsamples	0.977	0.020
Behavioural	Ward 80% subsamples	0.607	0.088
Combined	K-Means seed runs	0.987	0.008
Combined	K-Means 80% subsamples	0.976	0.015
Combined	Ward 80% subsamples	0.738	0.065

These results indicate that behavioural and combined K-Means partitions were reproducible under the tested perturbations, while demographic K-Means was more sensitive to initialisation. High ARI should not be interpreted as proof that a partition is substantively correct; it indicates repeatability relative to the tested procedure.

4.3 Agreement across representations

K-Means demographic and K-Means behavioural partitions had ARI 0.276 and NMI 0.274 which shows a small amount of agreement. K-Means demographic and K-Means combined partitions had ARI 0.309. K-Means behavioural and K-Means combined partitions were more alike with ARI 0.873 and NMI 0.808. These findings suggest that adding K-Means variables, to the K-Means behavioural feature set did not change the K-Means grouping dramatically while K-Means demographic-only grouping captured a different structure.

Pair of representations	Algorithm	ARI	NMI
Demographic vs Behavioural	K-Means	0.276	0.274
Demographic vs Behavioural	Ward	0.097	0.100
Demographic vs Combined	K-Means	0.309	0.322
Demographic vs Combined	Ward	0.107	0.115
Behavioural vs Combined	K-Means	0.873	0.808
Behavioural vs Combined	Ward	0.545	0.547

4.4 External validation

The overall campaign response rate in the cleaned sample was 14.87 percent. Demographic segmentation showed the widest cluster-level response spread: 7.81 percent to 26.47 percent a difference of 18.66 percentage points and Cramérs V of 0.218. Behavioural segmentation showed response rates from 8.62 percent to 22.08 percent, a spread of 13.46 points and Cramérs V of 0.165. Combined segmentation showed response rates from 9.02 percent to 23.98 percent, a spread of 14.95 points and Cramérs V of 0.178. The reported chi-square p-values were very small, for all three. Statistical significance alone does not establish practical value.

Representation	Minimum response	Maximum response	Spread (pp)	Cramér's V	Eta²: spend
Demographic	7.81%	26.47%	18.66	0.218	0.523
Behavioural	8.62%	22.08%	13.46	0.165	0.675
Combined	9.02%	23.98%	14.95	0.178	0.685

Behavioural and combined segmentations showed a difference in total spending compared to demographic segmentation. The eta-squared values were 0.675 for behavioural and 0.685 for combined while, for demographic it was 0.523. This makes sense because spending-related features are already included in the combined clustering methods. The external response outcome offers an independent test of the models even though campaign response might still connect to past spending habits.

4.5 Cluster profiles and practical meaning

The behavioural K-Means profiles suggest three groups. Cluster 1 (975 customers; 43.7 %) is spending, with average total spend about 86.8 and relatively low channel purchase counts; its response rate is 8.62 %. Cluster 2 (555; 24.9 %) has mid-level spend (about 725.2) deal purchases (4.27) and web purchases (6.82); its response rate is 16.76 %. Cluster 3 (702; 31.5 %) has the average total spend (about 1,231.7) high store and catalogue purchases and the highest response rate (22.08 %). These are profiles, not permanent customer identities.

For a marketing team Cluster 1 could be tested with low-cost reactivation messages, relevant product bundles or channel-friction checks of expensive blanket discounts. Cluster 2 could be tested with targeted promotions and cross-channel journeys while monitoring margin because deal purchases are relatively high. Cluster 3 could receive retention, loyalty and cross-category recommendations while ensuring offers do not unnecessarily discount purchases customers would make anyway. A controlled campaign experiment is needed before claiming that these actions improve response or profitability.

Demographic clusters show a managerial use. One demographic cluster has higher average income (about 76,003) and higher average spend (, about 1,296.7) while the two lower-income clusters differ

in partnership status and household children. Since demographic response spread is highest demographic features may help identify campaign response differences but these segments should be used carefully and not as labels. The combined segmentation largely resembles K-Means (ARI 0.873) and may provide added context for customer descriptions.

4.6 Robustness

Several robustness checks support the K-Means solution. When duplicate records were removed, ignoring ID the ARI was 0.956 and silhouette was 0.320. Dropping rows with missing income gave an ARI of 0.997 and silhouette of 0.319. Using PCA to retain 95% of variance led to an ARI of 0.999 and silhouette of 0.325.. When total spend was replaced with six separate log-transformed spend-category features the ARI dropped to 0.733 and silhouette to 0.276. This shows the segmentation is sensitive to how spending's represented. Removing education from features gave ARI 0.916 and silhouette 0.282 compared to the main demographic solution. These checks should be called sensitivity evidence, not proof that the model's universally robust.

4.7 Statistical and practical interpretation

The external-validation tests show small chi-square p-values for the link between segmentation and response. With than two thousand observations small differences can be found to be statistically significant; therefore effect sizes and the response-rate spread should be looked at together with p-values. Cramérs V is not very high for all three representations, with the demographic representation being the highest, at 0.218. This does not mean that demographic targeting is always the choice: the highest response rate is one part of the decision while campaign costs the number of people reached, extra increase and profit are also important. Similarly eta-squared for spending is partly affected by using spending-related features to create combined clusters so it should not be shown as proof of how well the model works on its own. A practical next step is to set up a pilot of using the labels right away. The marketing team could use the K-Means clusters to divide eligible customers randomly put customers within each group into a treatment and a control and check the extra conversions, contribution margin, average order value and the rate of complaints or unsubscribes. Results should be shared with confidence intervals. Checked for business importance. Segment labels should only be updated through a process so that campaign results stay the same over time.

4.8 Cross-criterion interpretation

The results show why choosing a segmentation should be treated as a decision problem, not a leaderboard competition. Behavioural clustering got the internal scores meaning spending and channel behaviour created a clearer separation after standardisation. Ward clustering on features had a silhouette of 0.324 and Davies–Bouldin index of 1.236. Behavioural K-Means had a silhouette of 0.319. The small difference in silhouette scores doesn’t pick a winner. The K-Means behavioural partition showed

subsample stability in the tests. Also K-Means is simpler to use for assigning customers when the model is refitted or centroids are kept.

External response results give a ranking. Demographic segmentation had the response-rate spread. 18.66 Percentage points. Compared to 13.46 for behavioural segmentation and 14.95 for the combined representation. This does not contradict the internal quality of behavioural clustering. Internal metrics ask if the data points group well in the input space. Response-rate spread asks whether the groups differ on a campaign outcome. A feature set can do well on one measure and less well on another. The demographic result might help generate ideas. It shouldn't be taken as proof that demographics cause response or that targeting by demographics is always the most profitable.

The spending results need care. Behavioural and combined clusters are built from spending data so high eta-squared values for spending are expected. They shouldn't be treated as validation. Campaign response is more external to the clustering features though the data still reflect past marketing choices and customer selection. Even if response rates differ statistically real decisions must consider uncertainty, contact costs, incremental margin, customer fatigue and whether past campaigns were assigned randomly.

Cluster sizes matter too. The behavioural K-Means solution created groups of 975, 555 and 702 customers. This avoided having a small segment, which happened in the main matched-k comparison. The behavioural Ward solution had a cluster of about 9.4%, which might still be useful but could need more careful handling. A small cluster can represent a niche but it can also be unstable or costly to target. The best choice depends on campaign economics, minimum audience size, the offer available and whether the group's traits make sense to marketing people.

Finally stability results depend on the chosen features, scaling and k. Behavioural K-Means had subsample stability. Mean ARI 0.977. While Ward's was lower at 0.607 under the same resampling setup. The combined K-Means solution was very stable across seeds and subsamples but its internal quality was weaker, than behavioural clustering. So stability alone isn't enough. The best operational option should balance geometry, stability, interpretation, response differentiation and implementation cost. It should also be tested in a controlled campaign to validate it prospectively.

5. Discussion

The main finding shows that the behavioural representation led to the internal cluster quality when the data was split into three clusters. It also showed stability in K-Means subsampling and clear differences in spending and campaign response. This supports using behaviour as the main basis for segmentation when the goal is to separate customers by purchase intensity, channel preference and sensitivity to deals. This result matches research that emphasizes customer representation and feature choice as key to segmentation rather than relying on the algorithm alone to decide quality (Wedel & Kamakura 2000; Hutter et al., 2023).

Demographic representation should not be ignored. It had the spread in response rates and the clusters reflected differences in income, partnership status and whether households had children. This suggests that the usefulness of a segment depends on the decision objective. Behavioural features worked best for cluster structure and spending differences. Demographics showed variation in response rates. The combined representation had the eta-squared for spending but that result isn't independent because spending-related features were part of the input data.

K-Means and Ward clustering did not produce the groupings. The behavioural Ward method had better silhouette scores and lower Davies–Bouldin values than behavioural K-Means.. Its smallest cluster was only 9.4% of the sample. This shows why relying on one metric is not enough. A small cluster might represent a niche or a difficult-to-serve group. It needs profiling and input from stakeholders. K-Means is a primary model here because its behavioural clusters are stable and more evenly distributed.. The final choice should be seen as a trade-off, not a clear winner.

The results are descriptive only. The campaign response variable reflects responses and does not prove that segment labels cause those responses. Also the data's collection date, source and business context are not included in the files, which limits how well the results can be applied outside this setting. Using a latest-customer-date proxy, for age and tenure encoding of education assuming "2n Cycle" and "Master" are equivalent and removing extreme values should be clearly explained. Future work should test the segments on a later time period assess profitability and campaign costs and explore distance measures or mixed-data clustering methods.

6. Managerial Implications

- Use K-Means as the first operational segmentation candidate when the goal is to customize marketing around spending intensity and channel behaviour. Validate the chosen segment labels with marketing stakeholders.
- Use the behavioural segment for retention, loyalty and relevant cross-category recommendations. Avoid discounts to customers who already purchase heavily.
- Use the mid-spend segment to test targeted offers and channel-specific journeys. Evaluate margin, not just response rate.
- Treat the segment as a diagnostic group. Test. Onboarding messages. Examine whether low spending reflects engagement, limited need or channel friction.
- Use information as context for understanding segments. Do not use it as a substitute for behaviour or as a basis for unsupported assumptions about individual customers.
- Before rollout run a controlled experiment with holdout groups. Measure conversion, revenue, contribution margin, unsubscribe rates, complaint rates and campaign cost.
- Monitor segment stability over time. Re-estimate or review segments, on a planned cadence. Document how customers move between groups.

7. Research Contribution

The study offers a comparison of how demographic behavioural and combined customer representations perform under the same three-cluster setup. It uses two algorithms and several validation methods. The key finding is that the best representation depends on what you're trying to achieve. Behavioural features work best when you want to separate customers based on spending and internal quality. Demographic features tend to give the range of responses in marketing campaigns. Combining both types of data with K-Means mostly gives results to using behavioural data alone. The analysis also shows that a reliable approach must include stability checks and sensitivity testing—not one single clustering score.

The contribution is an applied -criterion comparison of customer representations using the given dataset. It should not be taken as proof that this method's better, in all cases. The choice of representation depends on the decision goal. Limitations remain, including the dataset’s origin. How well the results apply over time.

8. Limitations and Future Research

First dataset provenance and collection date are not documented in the files provided. Second the data seem to represent one customer base and one historical campaign context so findings might not apply to retailers, other countries or other periods. Third age and tenure are based on the recorded customer date as a proxy. Fourth education is ordinally. The idea that "2n Cycle" and "Master" are the same is an unverified assumption. Fifth Euclidean distance after standardisation might not fully show the differences in categorical and numeric data. Sixth the number of clusters and the way features are represented can affect the results. Seventh external validation uses outcomes and cannot prove cause and effect. Finally the literature includes segmentation reviews and a similar RFM/K-Means application but these sources do not fix the problem of verifying the dataset provenance. The comparison of the literature is not complete.

Future research should check where the data comes from and use a holdout period that is separated in time; look at profitability. How much more campaigns improve results; test different ways to measure distance and different algorithms for mixed data; check for fairness and unfair effects when needed; and get opinions, from marketing experts through a qualitative review. A next experiment could give customers in each group campaign treatments and a group that does not get any campaign to see the real effect of the campaigns.

9. Conclusion

This study looked at three types of customer segmentation: one based on demographics, one based on behavior and one that combined both. They used two methods K-Means and Ward clustering. At three clusters the behavior-based approach had the internal quality. The behavior-based K-Means method was very stable when tested with samples. The demographic-based approach had the range in response rates. The behavior-based and combined approaches showed differences in spending. The main takeaway is to start with behavior-based segments for work and use demographic details as extra information. Actions for each segment should be tested through experiments. The findings are specific, to the dataset. Should be seen as descriptions of that data. Before handing it in the group needs to make sure where the dataset came from add the three papers finish the full review of existing research and check that all results can be reproduced from the notebook.

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